

PN JUNCTION AT EQUILIBRIUM

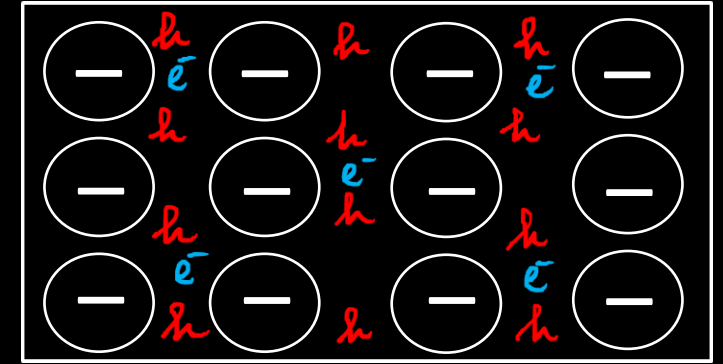
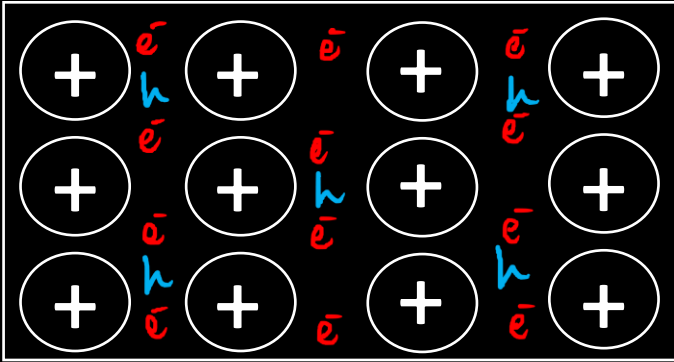
DAY - 03 | [23-03-2025]

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[Spec. Microelectronics & VLSI]

Representation of N – TYPE s/c $\rightarrow$ EXTRINSIC

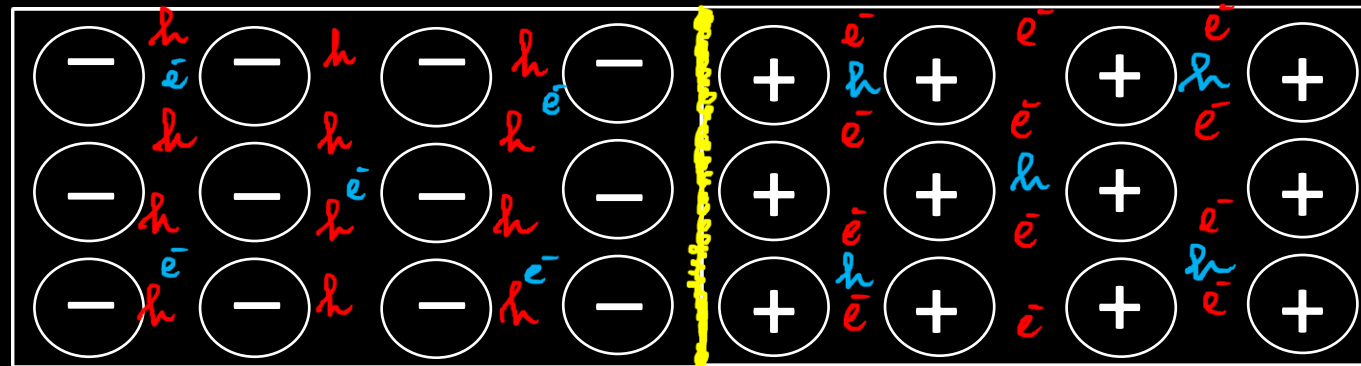
EXTRINSIC $\leftarrow$ Representation of P – TYPE s/c



+ = Immobile DONOR ions .

- = Immobile ACCEPTOR ions .

e-s & h = MOBILE / CHARGE CARRIERS =



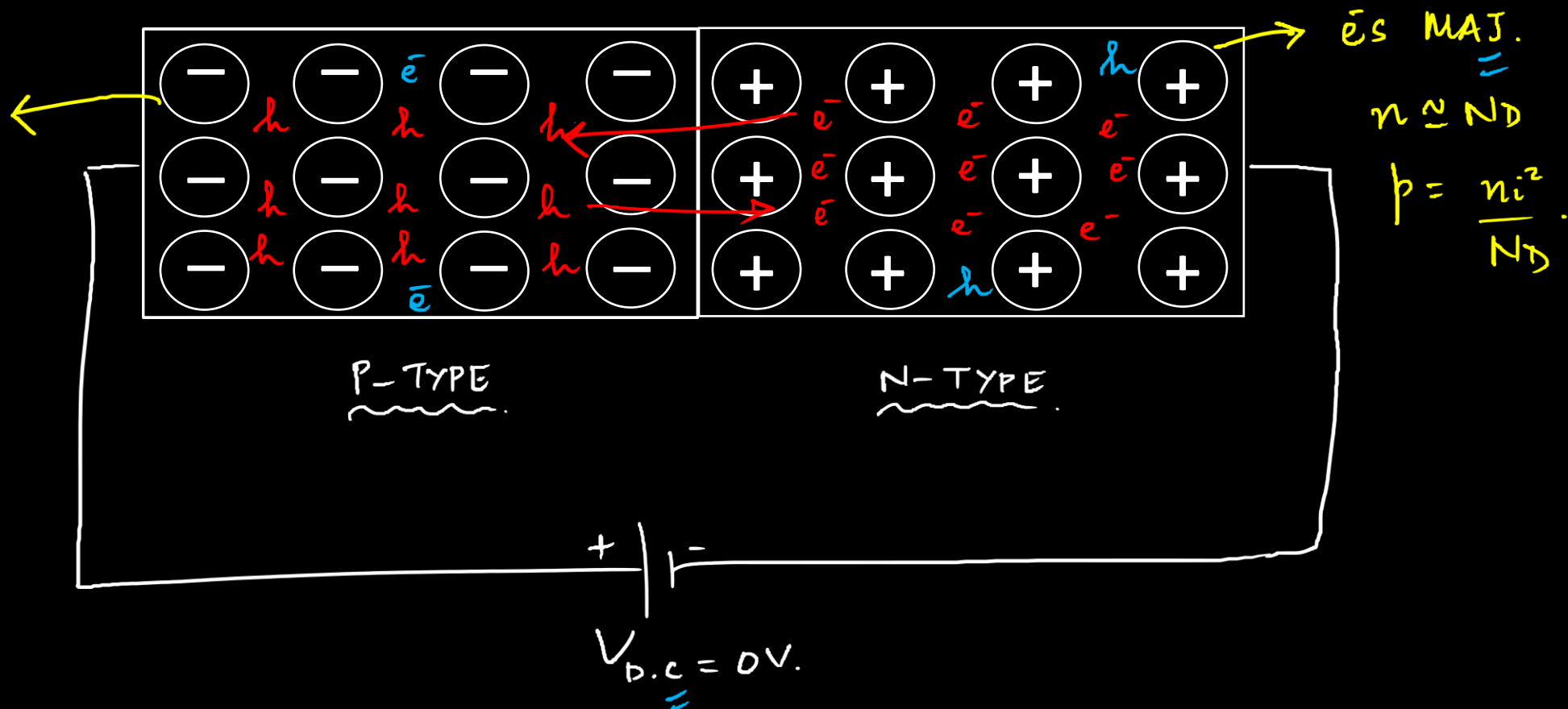
$\downarrow$ METALLURGICAL JUNCTION =
P – N Junction

$\rightarrow$ The interface joining /
Separating the P-TYPE &
N-TYPE extr. s/c is k/a/s ...

HOLES MAJ. =

$$p \approx N_A$$

$$n = \frac{n_i^2}{N_A}$$



P-N Junction at Equilibrium $\rightarrow \bar{N}_A$ and $\bar{N}_D$ conc $\rightarrow$ UNIFORM throughout the Semiconductor =

According to Material Science,

holes from p-side will diffuse to n-side

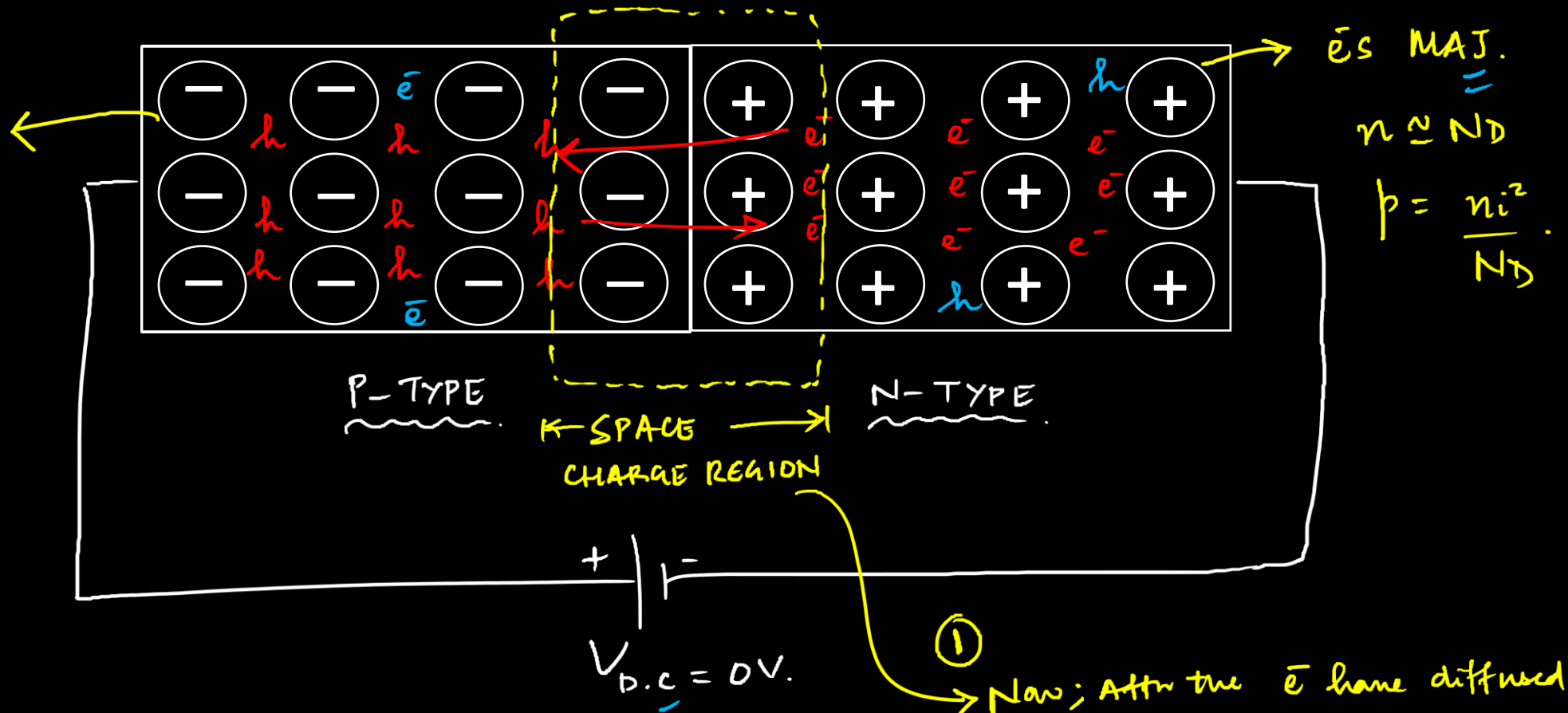
e^-

$n \rightarrow p.$

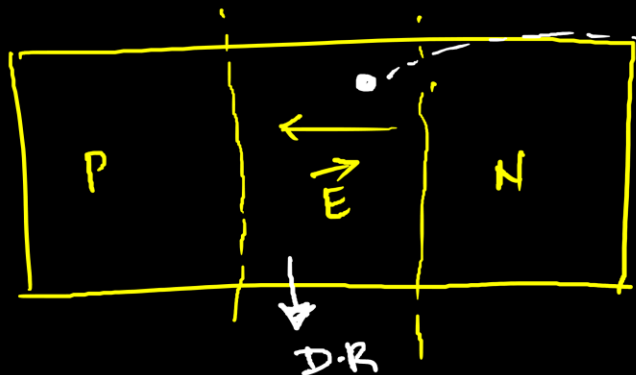
HOLES MAJ.

$$p \approx N_A$$

$$n = \frac{n_i^2}{N_A}$$



② So, An $\vec{E}$ is induced from N $\rightarrow$ P region.



why the "D.R." is devoid of CHARGE CARRIERS ??

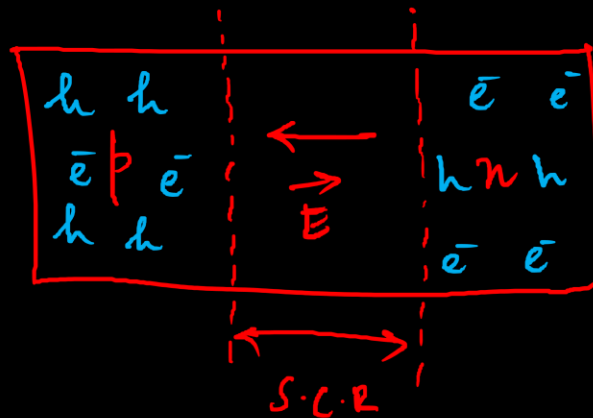
① Now; After the e^- have diffused from N $\rightarrow$ P side and so the holes from P $\rightarrow$ N side; A NET POSITIVE & NEGATIVE immobile charge induces.

N-TYPE $\rightarrow$ higher potential.
P-TYPE $\rightarrow$ lower potential.

The Force Applied By the $\vec{E}$ to the:

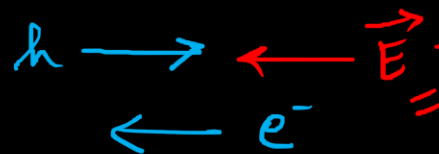
(i) $\bar{e}_s$:

$$\vec{F} = z_e \vec{E}$$

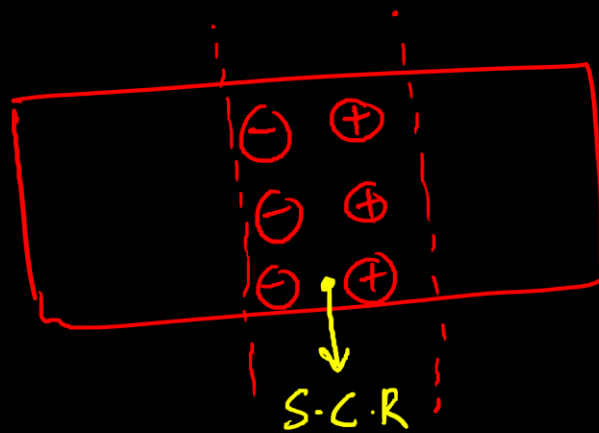


(ii) h_{oh} :

$$\begin{aligned} \vec{F} &= z_p (-\vec{E}) \\ &= -\vec{E} \cdot z_h \end{aligned}$$



→ This $\vec{E}$ is helping the $\bar{e}$'s diffusion from $N \rightarrow P$ side.



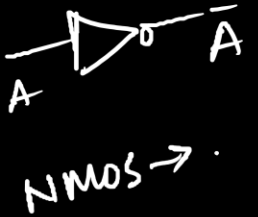
Under Thermal Equilibrium;

$$\begin{aligned} \text{DIFFUSION FORCE} &= \vec{E} \text{ due to NET IMMOBILE ions} \\ &N_A^- \text{ \& } N_D^+ \\ &\ominus \quad \quad \oplus \end{aligned}$$

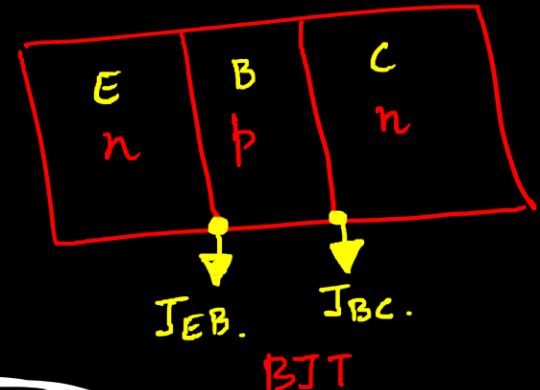
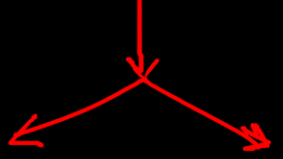
$\left. \begin{array}{l} N_A \checkmark \\ N_D \checkmark \end{array} \right\} \rightarrow \text{DIFFUSION due to diff. Immobile ions conc.}$
 $= \rightarrow \text{Diffusion force due to DENSITY!! gradient!!}$

①. TRANSISTOR

Transfer + Resistor =



PLACEMENTS



i/p - HIGH impedance → NOISE → MOSFET.
 i① → ✓
 MARGIN =

- 1s-2s Complement ✓
- 4:1 MUX ✓
- op-amp

AEC → D.C / A.C response
 B29
 92B
 voltage feedback

FFS

Latches
 Registers
 Counters.

HDL

Verilog / JAVA / Python =
 Non-Blocking / Blocking =

② DOPING: E > C > B

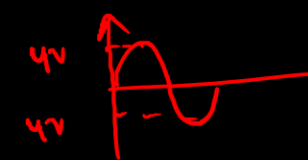
SIZE: C > E > B.

$I_C = \beta \cdot I_B$
 mA
 $I_E \rightarrow mA$

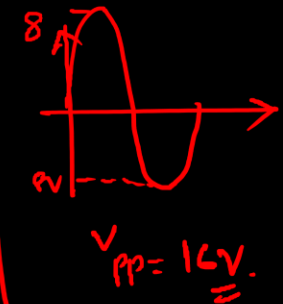
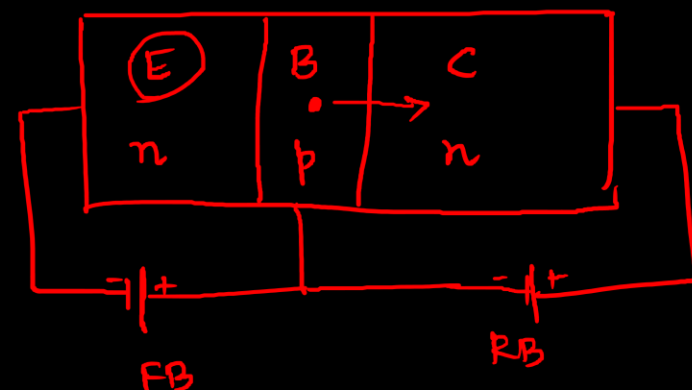
③ CC, CE, CB
 Ampl.

CE - CC - CB =

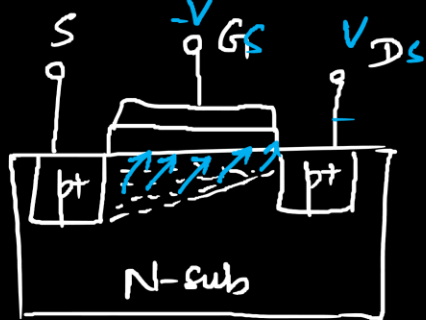
$V_{P-P} = 8V$



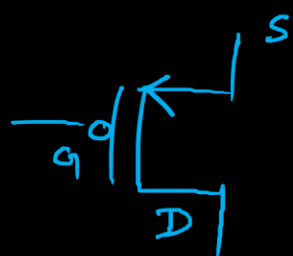
A.VLSI → CS → Current Buffer ✓



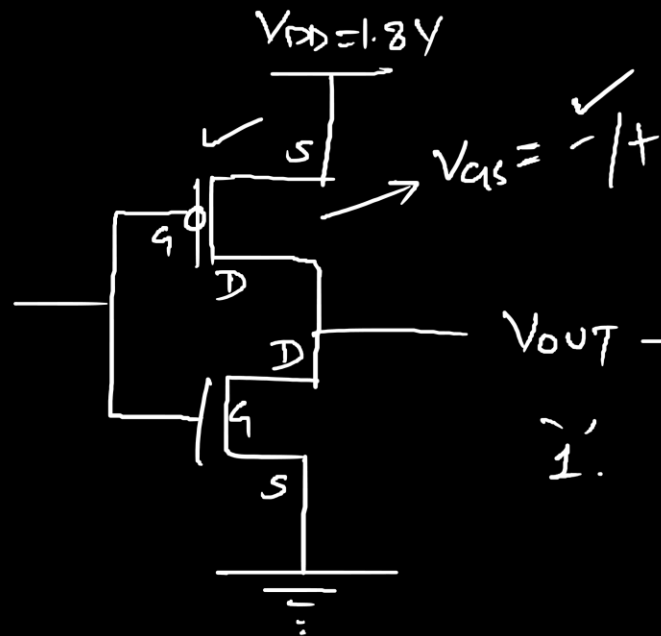
→ Amplifier



P-MOS.



V_{IN}
0



BUFFER.

